

# Omni-Context: Evidence-Linked Manuscript Scaffold

[TODO: Author names and affiliations required.]

## Abstract

[TODO: Author prose required.]

## 1 Introduction

[CITATION NEEDED: long-term memory for language agents] [TODO: Author prose required.]

## 2 Related Work

[CITATION NEEDED: retrieval-augmented generation and agent memory] [CITATION NEEDED: temporal knowledge representation] [CITATION NEEDED: evaluation of long-horizon language agents] [TODO: Author prose required after literature verification.]

## 3 Method

[TODO: Author prose required.]

## 4 Benchmark

[TODO: Author prose required.]

## 5 Experimental Setup

[TODO: Author prose required.]

**External evaluation protocol.** We prepared fixture-tested, phase-separated adapters and preregistrations for LongMemEval-S cleaned and LoCoMo question answering. Generation cannot open Gold, and formal execution requires a custodian authorization bound to exact dataset, product, adapter, and preregistration hashes. This paper asset preparation did not execute either formal held-out benchmark.

## 6 Results

[TODO: Author interpretation required.]

## 7 Analysis

[TODO: Author analysis required.]

## 8 Limitations

[TODO: Author prose required; do not remove any verified limitation above.]

External validity remains unestablished: LongMemEval is NOT RUN, and LoCoMo Conversations 2–10 remain unaccessed. LongMemEval-V2 requires trajectory, environment-state, and multimodal interfaces absent from the frozen product and is therefore follow-up work rather than a result of this study.

## 9 Conclusion

[TODO: Author prose required.]

## Omni-Context frozen architecture and evaluation boundary

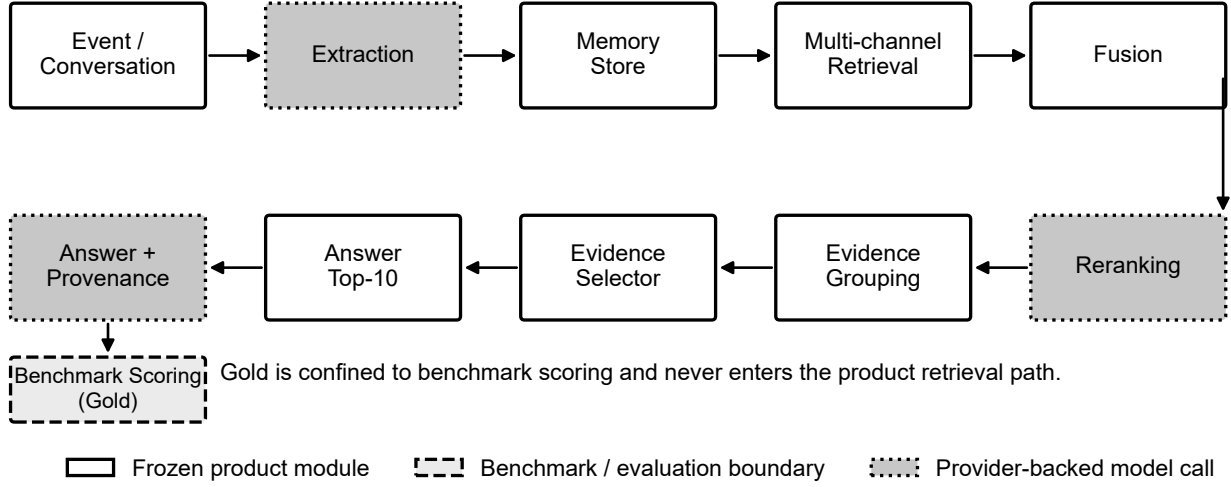


Figure 1: Frozen product architecture and evaluation boundary. Product modules, benchmark boundaries, and provider-backed calls use distinct line/fill styles.

Table 1: Version and reproducibility locks.

| Item                  | Locked value                                                                                     |
|-----------------------|--------------------------------------------------------------------------------------------------|
| Product commit        | 17dc1d0107b0474de84058205a91b302ba290a74                                                         |
| Benchmark commit      | 62b0b20f944f7e9a2c58f02ce1c65bb43dfbf841                                                         |
| Evidence commit       | ad1fe8806255e420e65398ae67df0a50474356d4                                                         |
| Embedding             | Xenova/multilingual-e5-large, revision a19b072cb4f0cc8bf98b4e46f90a787a61380979, 1024 dimensions |
| Answer                | deepseek-v4-flash, max tokens 1200, temperature 0, thinking disabled                             |
| Deterministic rescore | 105 records, 0 differences, 0 scoring defects                                                    |
| Open issues           | P0=0; P1 classes=2                                                                               |
| LoCoMo                | NOT RUN; Conversation 2–10 accessed=false                                                        |

## Frozen retrieval evidence pipeline

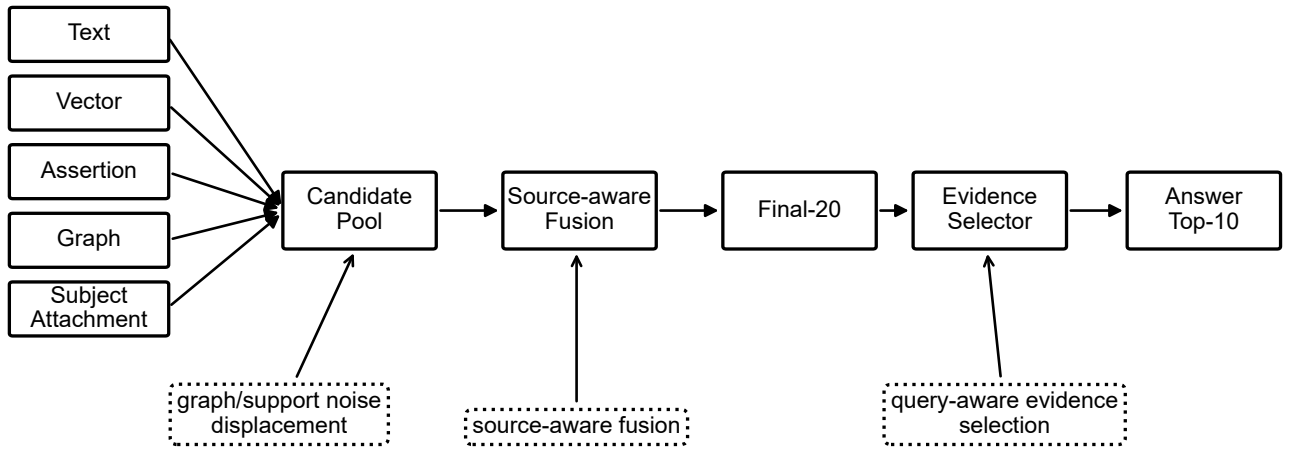


Figure 2: Evidence retrieval pipeline and the bounded scope of the final Candidate v3.1 fix.

Table 2: Version-locked internal benchmark results. Completed and error counts are never suppressed.

| Evaluation     | Mode           | Completed | Errors | Overall  |
|----------------|----------------|-----------|--------|----------|
| Targeted-7     | Full Omni      | 7/7       | 0      | 0.840868 |
| Development-35 | Full Omni      | 35/35     | 0      | 0.884842 |
| Development-35 | Retrieval-only | 35/35     | 0      | 0.563853 |
| Development-35 | No Memory      | 35/35     | 0      | 0.366713 |
| Formal-250     | Full Omni      | 248/250   | 2      | 0.866546 |
| Comparison-70  | Full Omni      | 70/70     | 0      | 0.870698 |
| Comparison-70  | Retrieval-only | 69/70     | 1      | 0.552346 |
| Comparison-70  | No Memory      | 70/70     | 0      | 0.364683 |

Formal-250 Overall is computed over successfully completed records using the benchmark’s category-macro definition. These mode comparisons are not strict component ablations. All evaluations shown here are internal Synthetic Curated Benchmark results; external held-out validity has not been established.

Table 3: Category-macro results. The mode delta is Comparison-70 Full Omni minus the stronger of Retrieval-only and No Memory.

| Category              | Dev Full | Formal Full | Comp Full | Comp Ret. | Comp No Mem. | $\Delta$ |
|-----------------------|----------|-------------|-----------|-----------|--------------|----------|
| Cognitive Continuity  | 0.916667 | 0.936667    | 0.956667  | 0.616667  | 0.500000     | 0.340000 |
| Memory Evolution      | 0.856746 | 0.852866    | 0.897963  | 0.407130  | 0.208333     | 0.490833 |
| Conflict Resolution   | 0.833333 | 0.812500    | 0.788889  | 0.327160  | 0.222222     | 0.461729 |
| Cross-Agent Transfer  | 1.000000 | 1.000000    | 0.994444  | 0.755556  | 0.500000     | 0.238888 |
| Human-like Forgetting | 0.927778 | 0.884722    | 0.886111  | 0.669444  | 0.416667     | 0.216667 |
| Proactive Insight     | 0.875000 | 0.770833    | 0.800000  | 0.487500  | 0.400000     | 0.312500 |
| Decision Quality      | 0.784370 | 0.808235    | 0.770815  | 0.602963  | 0.305556     | 0.167852 |

Values are benchmark category means, not scenario-weighted averages. Development and Formal baseline-by-category values are not invented; only available verified columns are shown.

Table 4: External held-out evaluation status. No score is available or implied.

| Benchmark                     | Adapter fixtures            | Preregistration                               | Formal re-<br>sult |
|-------------------------------|-----------------------------|-----------------------------------------------|--------------------|
| LongMemEval-S cleaned         | PASS (12 fictional records) | Adapter locked; dataset authorization pending | <b>NOT RUN</b>     |
| LoCoMo QA, Conversations 2–10 | PASS (4 fictional QA forms) | Adapter locked; dataset authorization pending | <b>NOT RUN</b>     |

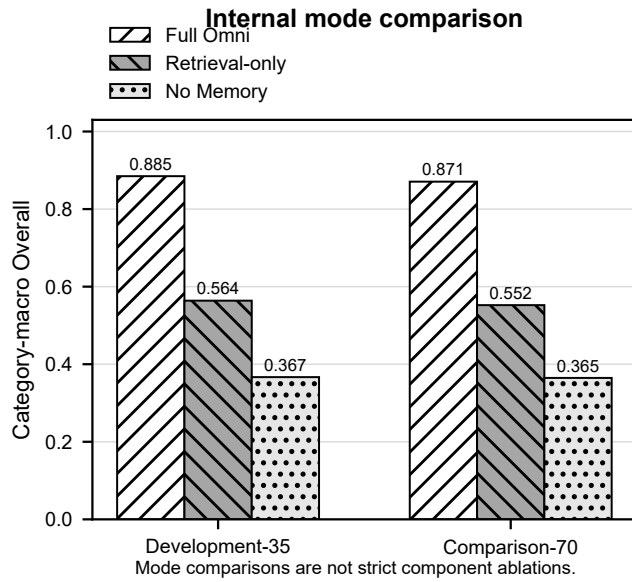


Figure 3: Internal Development-35 and Comparison-70 mode results.

Table 5: Efficiency and stability facts retained without error smoothing.

| Metric                  | Value     |
|-------------------------|-----------|
| Formal latency P50      | 150120 ms |
| Formal latency P95      | 243581 ms |
| Retry records           | 34        |
| Formal final errors     | 2         |
| Comparison final errors | 1         |

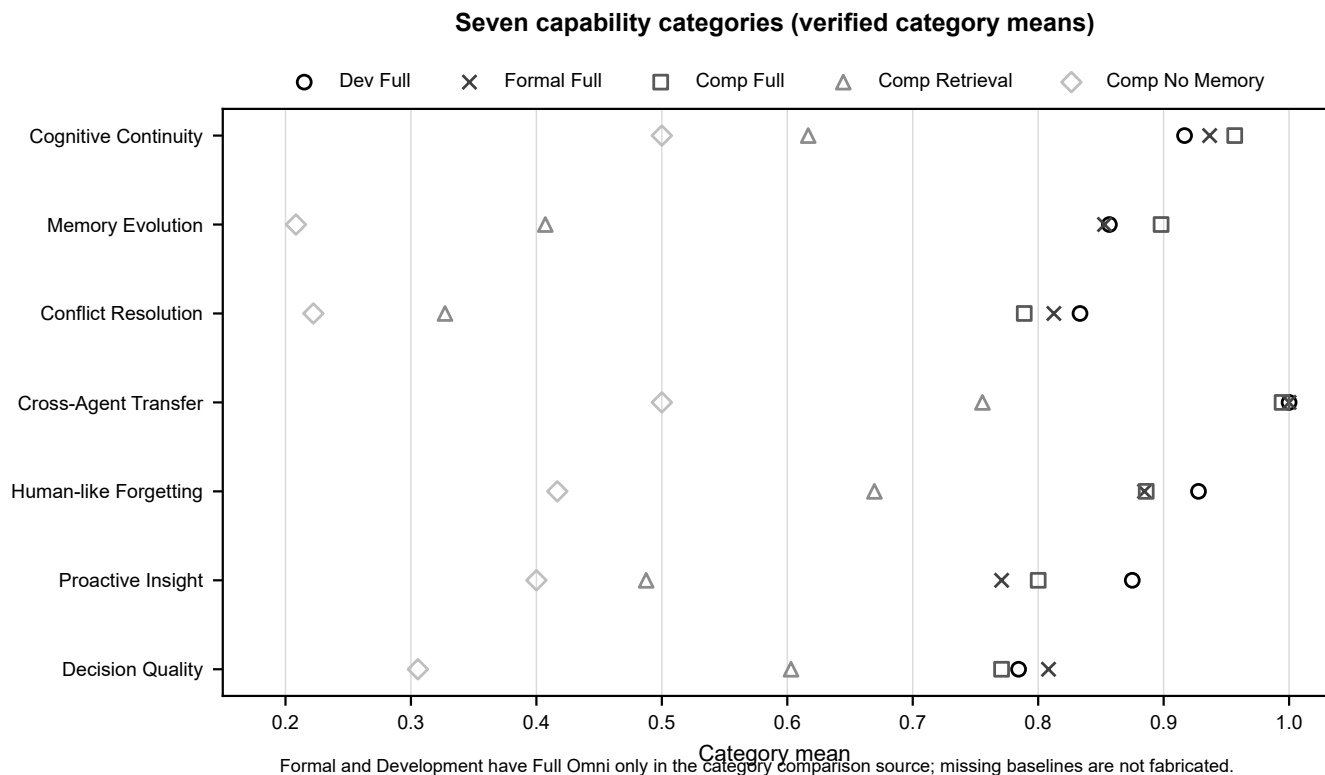


Figure 4: Verified category means. Missing Development/Formal baselines are not fabricated.

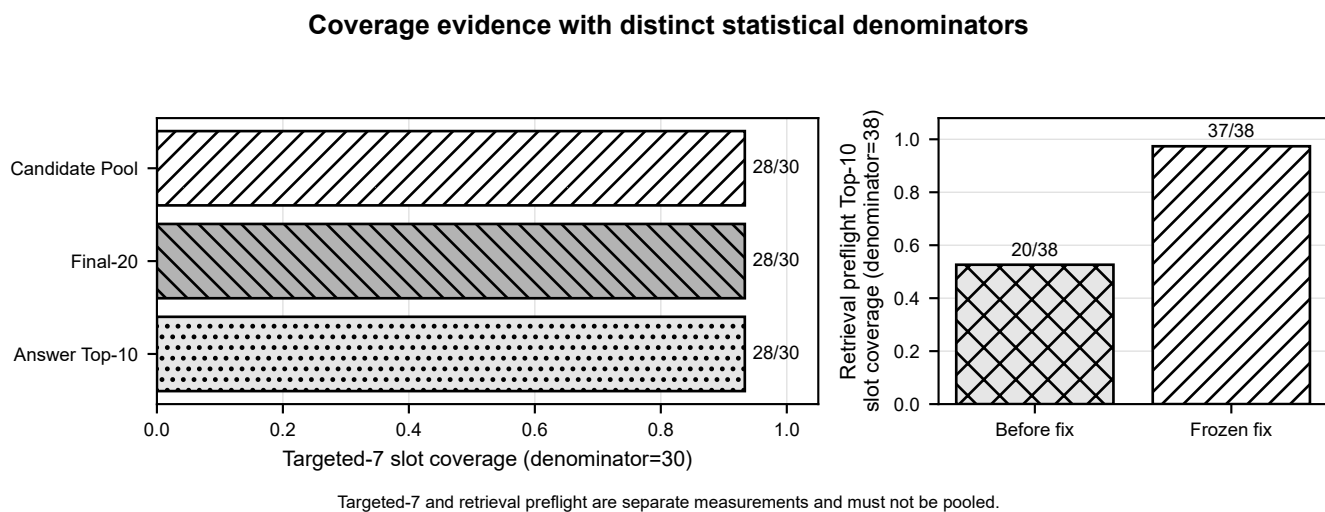


Figure 5: Coverage evidence with Targeted and retrieval-preflight denominators kept separate.

### Efficiency and provider/schema stability

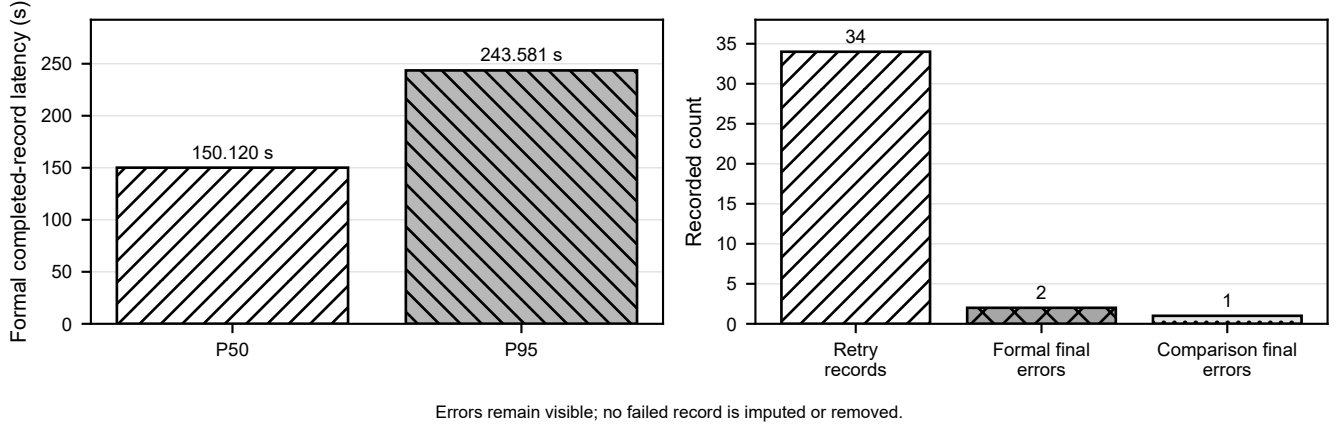


Figure 6: Formal latency and retained retry/error counts.

Table 6: Terminal errors retained after three prescribed attempts. No failed record entered score aggregation.

| Scenario                              | Failure                                                          | Attribution                              | Paper reporting                              |
|---------------------------------------|------------------------------------------------------------------|------------------------------------------|----------------------------------------------|
| formal-v2-memory_<br>evolution-039    | JSON truncation;<br>3 attempts; fin-<br>ish=length/length/length | Provider structured-output<br>truncation | Retained terminal error; no<br>score imputed |
| formal-v2-memory_<br>evolution-040    | JSON truncation;<br>3 attempts; fin-<br>ish=length/length/length | Provider structured-output<br>truncation | Retained terminal error; no<br>score imputed |
| formal-v2-conflict_<br>resolution-014 | empty source_ids;<br>3 attempts; fin-<br>ish=stop/stop/stop      | Baseline schema robustness               | Retained terminal error; no<br>score imputed |

Table 7: Development, Formal, and Comparison category distributions over completed records.

| Evaluation           | Category              | Completed | Errors | Median | Min   | Max   | Std. dev. |
|----------------------|-----------------------|-----------|--------|--------|-------|-------|-----------|
| Development Full     | Cognitive Continuity  | 5/5       | 0      | 1.000  | 0.750 | 1.000 | 0.105     |
| Development Full     | Memory Evolution      | 5/5       | 0      | 0.812  | 0.736 | 1.000 | 0.120     |
| Development Full     | Conflict Resolution   | 5/5       | 0      | 0.833  | 0.833 | 0.833 | 0.000     |
| Development Full     | Cross-Agent Transfer  | 5/5       | 0      | 1.000  | 1.000 | 1.000 | 0.000     |
| Development Full     | Human-like Forgetting | 5/5       | 0      | 1.000  | 0.694 | 1.000 | 0.119     |
| Development Full     | Proactive Insight     | 5/5       | 0      | 0.875  | 0.781 | 0.969 | 0.062     |
| Development Full     | Decision Quality      | 5/5       | 0      | 0.796  | 0.685 | 0.857 | 0.067     |
| Formal Full          | Cognitive Continuity  | 40/40     | 0      | 1.000  | 0.633 | 1.000 | 0.108     |
| Formal Full          | Memory Evolution      | 38/40     | 2      | 0.868  | 0.419 | 1.000 | 0.159     |
| Formal Full          | Conflict Resolution   | 40/40     | 0      | 0.833  | 0.389 | 0.833 | 0.080     |
| Formal Full          | Cross-Agent Transfer  | 30/30     | 0      | 1.000  | 1.000 | 1.000 | 0.000     |
| Formal Full          | Human-like Forgetting | 40/40     | 0      | 0.861  | 0.556 | 1.000 | 0.120     |
| Formal Full          | Proactive Insight     | 30/30     | 0      | 0.797  | 0.344 | 1.000 | 0.132     |
| Formal Full          | Decision Quality      | 30/30     | 0      | 0.815  | 0.602 | 0.926 | 0.067     |
| Comparison Full      | Cognitive Continuity  | 10/10     | 0      | 1.000  | 0.750 | 1.000 | 0.080     |
| Comparison Full      | Memory Evolution      | 10/10     | 0      | 1.000  | 0.631 | 1.000 | 0.133     |
| Comparison Full      | Conflict Resolution   | 10/10     | 0      | 0.833  | 0.389 | 0.833 | 0.133     |
| Comparison Full      | Cross-Agent Transfer  | 10/10     | 0      | 1.000  | 0.944 | 1.000 | 0.017     |
| Comparison Full      | Human-like Forgetting | 10/10     | 0      | 0.972  | 0.556 | 1.000 | 0.148     |
| Comparison Full      | Proactive Insight     | 10/10     | 0      | 0.781  | 0.688 | 0.938 | 0.104     |
| Comparison Full      | Decision Quality      | 10/10     | 0      | 0.787  | 0.537 | 0.880 | 0.102     |
| Comparison Retrieval | Cognitive Continuity  | 10/10     | 0      | 0.617  | 0.550 | 0.733 | 0.049     |
| Comparison Retrieval | Memory Evolution      | 10/10     | 0      | 0.384  | 0.208 | 0.648 | 0.201     |
| Comparison Retrieval | Conflict Resolution   | 9/10      | 1      | 0.222  | 0.222 | 0.611 | 0.160     |
| Comparison Retrieval | Cross-Agent Transfer  | 10/10     | 0      | 0.722  | 0.667 | 0.944 | 0.103     |
| Comparison Retrieval | Human-like Forgetting | 10/10     | 0      | 0.764  | 0.417 | 1.000 | 0.217     |
| Comparison Retrieval | Proactive Insight     | 10/10     | 0      | 0.484  | 0.281 | 0.812 | 0.150     |
| Comparison Retrieval | Decision Quality      | 10/10     | 0      | 0.606  | 0.481 | 0.727 | 0.078     |
| Comparison No Memory | Cognitive Continuity  | 10/10     | 0      | 0.500  | 0.500 | 0.500 | 0.000     |
| Comparison No Memory | Memory Evolution      | 10/10     | 0      | 0.208  | 0.208 | 0.208 | 0.000     |
| Comparison No Memory | Conflict Resolution   | 10/10     | 0      | 0.222  | 0.222 | 0.222 | 0.000     |
| Comparison No Memory | Cross-Agent Transfer  | 10/10     | 0      | 0.500  | 0.500 | 0.500 | 0.000     |
| Comparison No Memory | Human-like Forgetting | 10/10     | 0      | 0.417  | 0.417 | 0.417 | 0.000     |
| Comparison No Memory | Proactive Insight     | 10/10     | 0      | 0.406  | 0.375 | 0.438 | 0.023     |
| Comparison No Memory | Decision Quality      | 10/10     | 0      | 0.306  | 0.231 | 0.417 | 0.051     |

## A Detailed Category Evidence

Table 8: Verified category submetrics.

| Evaluation       | Category              | Submetrics                                                                                                                                                                                                        |
|------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Development Full | Cognitive Continuity  | profile recall=1.000; profile consistency=1.000; constraint utilization=0.667; personalization accuracy=0.833; contradiction rate=0.000; unsupported personalization rate=0.000                                   |
| Development Full | Memory Evolution      | current state accuracy=1.000; historical state preservation=1.000; temporal ordering accuracy=0.457; evolution interpretation accuracy=0.819; stale memory leakage=0.000; state transition accuracy=0.864         |
| Development Full | Conflict Resolution   | conflict resolution accuracy=1.000; latest valid fact accuracy=1.000; historical query accuracy=1.000; invalidated fact rejection=1.000; conflict disclosure accuracy=0.000; unsupported resolution rate=0.000    |
| Development Full | Cross-Agent Transfer  | cross agent recall=1.000; cross agent consistency=1.000; update propagation accuracy=1.000; provenance preservation=1.000; agent isolation error rate=0.000; stale transfer rate=0.000                            |
| Development Full | Human-like Forgetting | salient memory retention=1.000; noise suppression=0.933; false forgetting rate=0.000; stale retention rate=0.133; memory precision=0.967; invalidation accuracy=0.800                                             |
| Development Full | Proactive Insight     | insight precision=0.800; insight recall=0.750; blind spot detection rate=0.850; constraint awareness=0.900; actionability=0.750; unsupported claim rate=0.050; redundant insight rate=0.000; overreach rate=0.000 |

| Evaluation           | Category              | Submetrics                                                                                                                                                                                                                                        |
|----------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Development Full     | Decision Quality      | constraint coverage=0.900; goal alignment=0.750; personalization=0.733; option comparison quality=0.900; risk awareness=0.700; actionability=0.800; internal consistency=0.750; unsupported assumption rate=0.150; overall decision quality=0.676 |
| Formal Full          | Cognitive Continuity  | profile recall=0.930; profile consistency=1.000; constraint utilization=0.817; personalization accuracy=0.873; contradiction rate=0.000; unsupported personalization rate=0.000                                                                   |
| Formal Full          | Memory Evolution      | current state accuracy=0.947; historical state preservation=0.971; temporal ordering accuracy=0.525; evolution interpretation accuracy=0.814; stale memory leakage=0.000; state transition accuracy=0.861                                         |
| Formal Full          | Conflict Resolution   | conflict resolution accuracy=0.975; latest valid fact accuracy=0.975; historical query accuracy=0.950; invalidated fact rejection=0.950; conflict disclosure accuracy=0.025; unsupported resolution rate=0.000                                    |
| Formal Full          | Cross-Agent Transfer  | cross agent recall=1.000; cross agent consistency=1.000; update propagation accuracy=1.000; provenance preservation=1.000; agent isolation error rate=0.000; stale transfer rate=0.000                                                            |
| Formal Full          | Human-like Forgetting | salient memory retention=1.000; noise suppression=0.883; false forgetting rate=0.000; stale retention rate=0.142; memory precision=0.942; invalidation accuracy=0.625                                                                             |
| Formal Full          | Proactive Insight     | insight precision=0.750; insight recall=0.633; blind spot detection rate=0.742; constraint awareness=0.642; actionability=0.675; unsupported claim rate=0.083; redundant insight rate=0.117; overreach rate=0.075                                 |
| Formal Full          | Decision Quality      | constraint coverage=0.900; goal alignment=0.800; personalization=0.811; option comparison quality=0.833; risk awareness=0.642; actionability=0.758; internal consistency=0.850; unsupported assumption rate=0.050; overall decision quality=0.730 |
| Comparison Full      | Cognitive Continuity  | profile recall=0.960; profile consistency=1.000; constraint utilization=0.867; personalization accuracy=0.913; contradiction rate=0.000; unsupported personalization rate=0.000                                                                   |
| Comparison Full      | Memory Evolution      | current state accuracy=1.000; historical state preservation=0.960; temporal ordering accuracy=0.653; evolution interpretation accuracy=0.871; stale memory leakage=0.000; state transition accuracy=0.903                                         |
| Comparison Full      | Conflict Resolution   | conflict resolution accuracy=0.933; latest valid fact accuracy=0.900; historical query accuracy=0.900; invalidated fact rejection=0.900; conflict disclosure accuracy=0.100; unsupported resolution rate=0.000                                    |
| Comparison Full      | Cross-Agent Transfer  | cross agent recall=1.000; cross agent consistency=1.000; update propagation accuracy=1.000; provenance preservation=0.967; agent isolation error rate=0.000; stale transfer rate=0.000                                                            |
| Comparison Full      | Human-like Forgetting | salient memory retention=1.000; noise suppression=0.900; false forgetting rate=0.000; stale retention rate=0.133; memory precision=0.950; invalidation accuracy=0.600                                                                             |
| Comparison Full      | Proactive Insight     | insight precision=0.825; insight recall=0.675; blind spot detection rate=0.800; constraint awareness=0.650; actionability=0.700; unsupported claim rate=0.025; redundant insight rate=0.150; overreach rate=0.075                                 |
| Comparison Full      | Decision Quality      | constraint coverage=0.850; goal alignment=0.775; personalization=0.783; option comparison quality=0.775; risk awareness=0.675; actionability=0.725; internal consistency=0.825; unsupported assumption rate=0.150; overall decision quality=0.679 |
| Comparison Retrieval | Cognitive Continuity  | profile recall=0.300; profile consistency=1.000; constraint utilization=0.167; personalization accuracy=0.233; contradiction rate=0.000; unsupported personalization rate=0.000                                                                   |



| Evaluation           | Category              | Submetrics                                                                                                                                                                                                                                        |
|----------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Comparison Retrieval | Memory Evolution      | current state accuracy=0.500; historical state preservation=0.220; temporal ordering accuracy=0.033; evolution interpretation accuracy=0.251; stale memory leakage=0.000; state transition accuracy=0.438                                         |
| Comparison Retrieval | Conflict Resolution   | conflict resolution accuracy=0.407; latest valid fact accuracy=0.000; historical query accuracy=0.000; invalidated fact rejection=0.222; conflict disclosure accuracy=0.333; unsupported resolution rate=0.000                                    |
| Comparison Retrieval | Cross-Agent Transfer  | cross agent recall=0.500; cross agent consistency=1.000; update propagation accuracy=0.200; provenance preservation=0.833; agent isolation error rate=0.000; stale transfer rate=0.000                                                            |
| Comparison Retrieval | Human-like Forgetting | salient memory retention=0.600; noise suppression=0.967; false forgetting rate=0.400; stale retention rate=0.033; memory precision=0.783; invalidation accuracy=0.100                                                                             |
| Comparison Retrieval | Proactive Insight     | insight precision=0.400; insight recall=0.325; blind spot detection rate=0.325; constraint awareness=0.125; actionability=0.400; unsupported claim rate=0.325; redundant insight rate=0.050; overreach rate=0.300                                 |
| Comparison Retrieval | Decision Quality      | constraint coverage=0.850; goal alignment=0.575; personalization=0.667; option comparison quality=0.525; risk awareness=0.425; actionability=0.350; internal consistency=0.750; unsupported assumption rate=0.200; overall decision quality=0.485 |
| Comparison No Memory | Cognitive Continuity  | profile recall=0.000; profile consistency=1.000; constraint utilization=0.000; personalization accuracy=0.000; contradiction rate=0.000; unsupported personalization rate=0.000                                                                   |
| Comparison No Memory | Memory Evolution      | current state accuracy=0.000; historical state preservation=0.000; temporal ordering accuracy=0.000; evolution interpretation accuracy=0.000; stale memory leakage=0.000; state transition accuracy=0.250                                         |
| Comparison No Memory | Conflict Resolution   | conflict resolution accuracy=0.333; latest valid fact accuracy=0.000; historical query accuracy=0.000; invalidated fact rejection=0.000; conflict disclosure accuracy=0.000; unsupported resolution rate=0.000                                    |
| Comparison No Memory | Cross-Agent Transfer  | cross agent recall=0.000; cross agent consistency=1.000; update propagation accuracy=0.000; provenance preservation=0.000; agent isolation error rate=0.000; stale transfer rate=0.000                                                            |
| Comparison No Memory | Human-like Forgetting | salient memory retention=0.000; noise suppression=1.000; false forgetting rate=1.000; stale retention rate=0.000; memory precision=0.500; invalidation accuracy=0.000                                                                             |
| Comparison No Memory | Proactive Insight     | insight precision=0.000; insight recall=0.000; blind spot detection rate=0.000; constraint awareness=0.100; actionability=0.100; unsupported claim rate=0.000; redundant insight rate=0.000; overreach rate=0.000                                 |
| Comparison No Memory | Decision Quality      | constraint coverage=0.200; goal alignment=0.025; personalization=0.450; option comparison quality=0.000; risk awareness=0.050; actionability=0.175; internal consistency=0.800; unsupported assumption rate=0.000; overall decision quality=0.050 |

## References